Correlation Method for Detection of Transient T-Wave Alternans in Digital Holter ECG Recordings

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Background: Spectral and complex demodulation methods for detection of microvolt T-wave alternans (TWA) have limited ability to identify transient (nonstationary) TWA episodes. We aimed to develop and test new time-domain technique allowing TWA detection and quantification in as few as 7 consecutive beats from sinus rhythm ECGs.

Methods and Results: Quantification of TWA during sinus rhythm required preprocessing consisting of: low-pass filtering, RR stability testing, baseline and respiratory modulation removal, and T-wave windowing and synchronization. Our time-domain correlation method (CM) detects TWA by computing, for each consecutive T wave, an alternans correlation index based on a cross-correlation technique. CM allows quantitative analysis of the amplitude, duration, and overall magnitude of the TWA episode. The technical performance of CM was confirmed in testing with simulated TWA of varying amplitude, duration, and noisy conditions. The clinical performance of CM was demonstrated by analyzing digital Holter recordings of 39 long QT syndrome patients compared to 36 healthy subjects. CM identified TWA in 17 (44%) patients with nonstationary TWA detected in 8.

Conclusion: Our computer algorithms consisting of ECG preprocessing and TWA quantification by the correlation method provides the opportunity to detect nonstationary and stationary TWA in sinus rhythm of digital Holter ECG recordings. **A.N.E. 1999;4(4):416–424**

T-wave alternans; long QT syndrome; repolarization; signal processing; Holter monitoring

T-wave alternans (TWA), consisting of beat-to-beat alternation in the morphology of the T wave, is a marker of myocardial electrical instability.[1-4] Visible forms of TWA are associated with life-threatening ventricular arrhythmias instability.[2-4] In recent years, computerized analysis of digital electrocardiograms (ECGs) permitted the identification of microvolt, nonvisible TWA in experimental and clinical conditions.[5-8] Nearing et al.,[7] using a method based on complex demodulation, showed that microvolt TWA is associated with a decreased defibrillation threshold. Rosenbaum et al.,[8] using a spectral fast Fourier transformation analysis, showed that microvolt TWA is associated with an increased risk of ventricular arrhythmias. Both methods, but especially the spectral method, which is used in clinical studies, have limited ability to detect transient (nonstationary) episodes of TWA. ECG tracings with visible TWA show that TWA is a time-varying (nonstationary) phenomenon (Fig. 1), and therefore TWA detection algorithms should allow detection of transient (nonstationary) episodes of TWA.

In addition, microvolt TWA is usually detected in ECGs with pacing or exercise-induced tachycardia.[6,8-11] Although tachycardia increases the likelihood of TWA occurrence, TWA may also occur during sinus rhythm in resting conditions. Spontaneous sinus beat-to-beat RR variability can be a factor affecting the detection of TWA, and any

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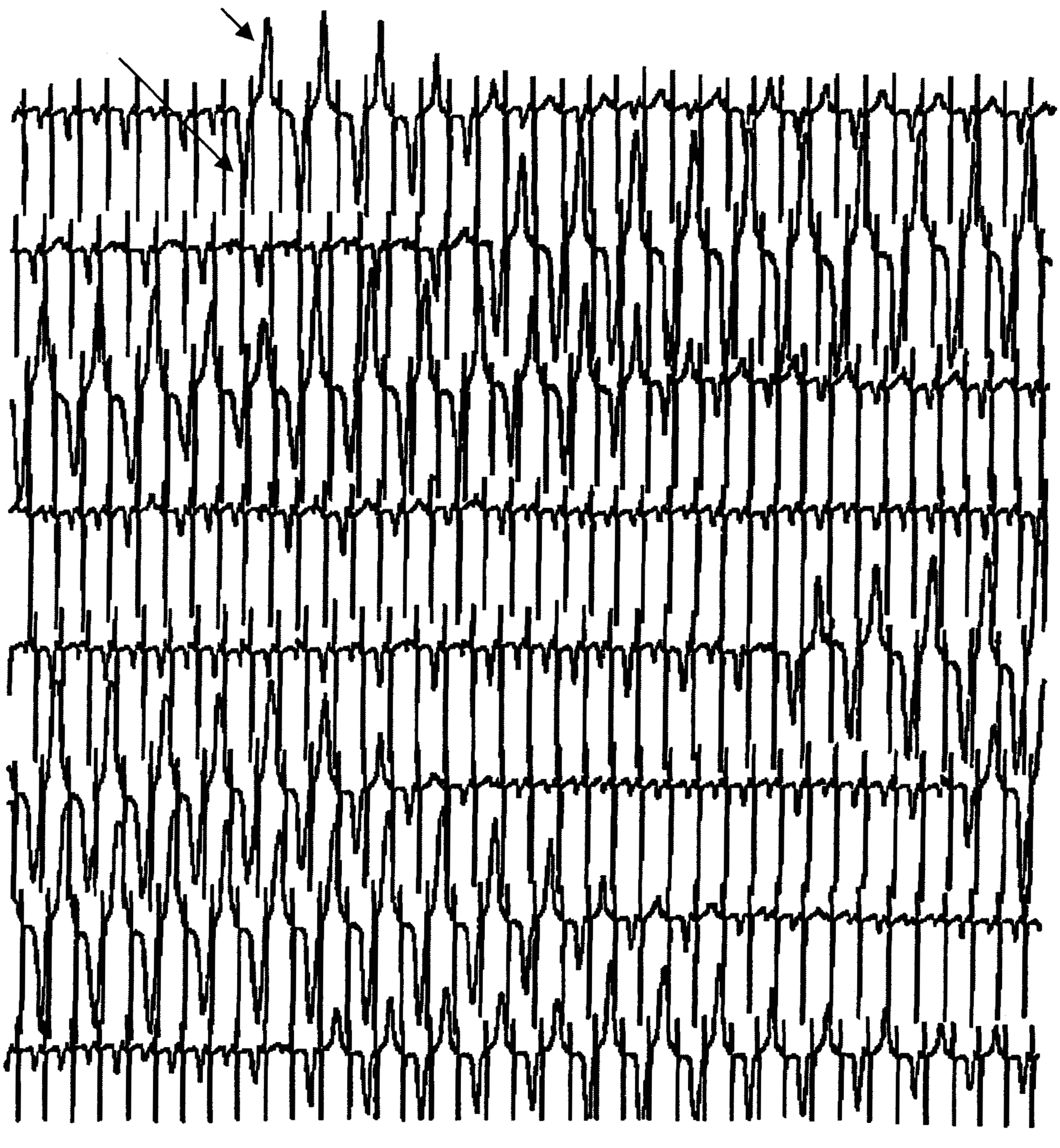

Figure 1. T-wave alternans episodes in a continuous Holter ECG recording in lead V_2 of a long QT syndrome patient. In every other beat, the changes in T-wave amplitude and morphology can be seen throughout the entire continuous recording (arrows indicate negative and positive T waves in two consecutive beats). This example illustrates a dynamic (nonstationary) nature of T-wave alternans.

technique analyzing TWA in sinus rhythm should account for RR variability. Given these clinically driven requirements, the primary aim of our study was to develop a new time-domain approach (the correlation method) for detecting and quantifying stationary (sustained) and nonstationary (transient) episodes of TWA in ECG recordings acquired during sinus rhythm. The secondary objective was to test the performance of our correlation method in a simulation study and in the clinical setting by analyzing digital Holter ECGs of long QT syndrome (LQTS) patients compared to healthy subjects.

METHODS

Signal-Processing Methodology

The algorithms described are designed for analysis of 128 consecutive beat ECGs from digital Holter recordings acquired with a sampling frequency of 1000 samples per second per channel. With some modification, the algorithms can also be used to identify TWA in ECG recordings of shorter length and lower sampling frequency.[12-14]

Preprocessing of ECG Data

The raw digital ECG data require preprocessing before being analyzed for TWA detection (Fig. 2). Initially, the ECG is low-pass filtered (cut-off frequency: 60 Hz) to reduce the background noise. The R peaks then are detected by analyzing the maximum derivative of the parabolic interpolation.[15] Next, RR stability is determined, and if the ECG has RR standard deviation > 10% mean RR, the ECG is rejected. Once the R peaks have been detected, each consecutive beat is analyzed to verify that it is a sinus beat (cross-correlation coefficient between beat and template > 0.8,[16] where the template is the median QRS complex from all 128 available QRS complexes). Variations in the morphology of the QRS complexes might occur as a consequence of arrhythmias and noise. If there are more than 10 nonsinus beats, the ECG is rejected. The ECGs with RR variability ≤ 10% and less than 10 nonsinus beats undergo baseline removal using a third-order spline interpolation of fiducial points in the PR interval.[15]

A windowing procedure is applied to the repolarization segment (if needed, the window can also encompass the QRS complex, allowing simultaneous QRS and T-wave alternans detection). For TWA detection purposes, the exact identification of the T-wave endpoints is not required. T waves are windowed using our heart rate-dependent formulae: for RR < 0.6, 0.6-1.1, and > 1.1 seconds, the onset of the T-wave window (W_{on}) is located 60, 100, and 150 ms after the R peak, respectively.[14] The length of the window (W_L) is also adjusted for heart rate with the empirical formula $W_L = 0.4\sqrt{mean\ RR}$.[14] Nonsinus beats have their

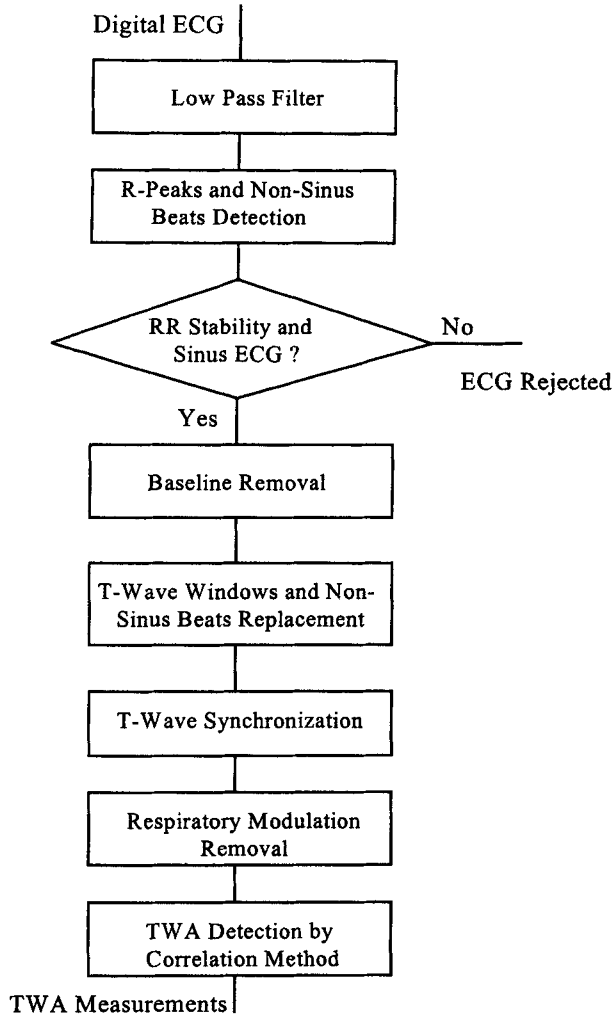

Figure 2. Schematic presentation of preprocessing steps required before TWA analysis in ECGs recorded during sinus rhythm.

repolarization segment replaced by the mean of the repolarization segments of the two previous sinus beats.

Synchronization (or alignment) of the T waves is accomplished using a cross-correlation technique. A reference T wave is calculated as the median of the 128 consecutive complexes using the initial positioning of the T-wave windows. Synchronization of the T wave is done keeping the length of the window constant, but letting its position vary from W_{on}-30 ms to W_{on}+30 ms, with increments of 1 ms. For each position of the T-wave window, the cross-correlation coefficient between the T wave and the reference T wave is calculated. Finally, the position of the T-wave window with the highest cross-correlation value is chosen.

After the T waves have been aligned, the final step in the preprocessing phase is to eliminate the variations of T-wave amplitude due to respiration. This operation is accomplished using a band-stop filter (cut-off frequencies: 0.14 and 0.35 cycles/beat) that eliminates frequencies in the respiratory range (typically 1/7-1/3 cycles/beat), because they do not overlap with the alternans frequency ($\frac{1}{2}$ cycles/beat, i.e., every other beat).

Time-Domain Correlation Method for TWA Detection

After preprocessing the raw ECG data, a median T wave (T_{mdn}) is computed from the 128 consecutive T waves. For each T_j wave, an alternans correlation index (ACI_j) is computed as:

$$ACI_j = \frac{\sum_{i=1}^{Ns} T_j(i) T_{mdn}(i)}{\sum_{i=1}^{Ns} T^2_{mdn}(i)} \qquad j = 1{:}128 \tag{1}$$

where N_s is the number of samples in each T-wave window.

ACI_j is a dimensionless quantity defined as the maximum value of the cross-correlation function between T_{mdn} and T_j, over the maximum value of the auto-correlation function of T_{mdn}. An $ACI_j > 1$ indicates that T_j is 'larger' than T_{mdn}, whereas an $ACI_j < 1$ indicates that T_j is 'smaller' than T_{mdn}. An $ACI_j < 0$ indicates that T_j and T_{mdn} have opposite polarity. In the presence of TWA, ACI_j as a function of j (j = 1,2. . .128) alternates around the value of 1 in the case of monophasic TWA (T wave does not change polarity; Fig. 3) or around the value of zero in the case of biphasic TWA (T wave changes polarity every other beat). T-wave alternans is detected when there are at least seven alternating beats.

Because a value of ACI_j is calculated for each T wave, it is possible to determine the duration of the TWA episode as the number of alternating beats (N_{CM}). Using T_{mdn} (in μV) and ACI_j, TWA amplitude (in μV) can be estimated as follows.

From Equation 1, assuming $ACI_j > 1$:

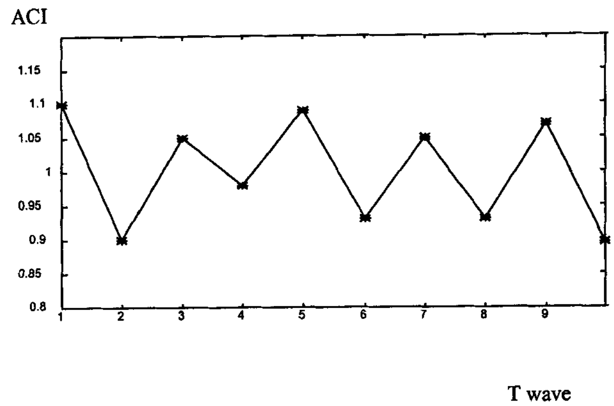

Figure 3. Alternans Correlation Index (ACI_j) for each of the consecutive T waves. Up-and-down trend of the plot indicates the presence of T-wave alternans.

$$ACI_j - 1 = \frac{\sum_{i=1}^{Ns} T_j(i)\, T_{mdn}(i)}{\sum_{i=1}^{Ns} T^2_{mdn}(i)} - \frac{\sum_{i=1}^{Ns} T^2_{mdn}(i)}{\sum_{i=1}^{Ns} T^2_{mdn}(i)} = \frac{\sum_{i=1}^{Ns} T_{mdn}(i)\, T_j(i) - T_{mdn}(i)}{\sum_{i=1}^{Ns} T^2_{mdn}(i)} \qquad j = 1{:}128 \tag{2}$$

Because it is not possible to know a priori what part of the T wave is alternating, we made the assumption, common to other TWA detection techniques,[6,8] that all samples of the same T wave are alternating the same quantity (in our case this assumption influences the quantification but not the detection of TWA). Consequently, $(T_j(i) - T_{mdn}(i)) = A_{CM}(j)/2$, where $A_{CM}(j)$ is a constant dependent of the beat j but not of the sample i. Thus, Equation 2 becomes:

$$ACI_j - 1 = \frac{\frac{A_{CM}(j)}{2} \sum_{i=1}^{Ns} T_{mdn}(i)}{\sum_{i=1}^{Ns} T^2_{mdn}(i)} \qquad j = 1{:}128 \tag{3}$$

In Equation 3, the factor '2' comes from the fact that the value of the alternans has to be measured between two consecutive waves and not between a T wave and T_{mdn} (Fig. 4).

From Equation 3:

$$A_{CM}(j) = 2\,ACI_j - 1\frac{\sum_{i=1}^{N_S} T^{e}_{mdn}(i)}{\sum_{i=1}^{N_S} T_{mdn}(i)} \quad j = 1{:}128 \qquad (4)$$

Because ACI_j can assume any value and the T wave can have positive or negative amplitudes, the final formula for estimating TWA amplitude ($A_{CM}(j)$) by the correlation method is:

$$A_{CM}(j) = 2|ACI_j - 1|\frac{\sum_{i=1}^{N_S} T^{e}_{mdn}(i)}{\sum_{i=1}^{N_S} |T_{mdn}(i)|} \quad j = 1{:}128 \qquad (5)$$

The values of $A_{CM}(j)$ are computed only for alternating T waves. For those beats not involved in a TWA episode $A_{CM}(j)$ is set at zero. Consequently, $A_{CM}(j)$ is greater or equal to zero, and expresses the amount of alternans (in μV) that characterizes each T_j wave. $A_{CM}(j)$ values greater than zero are averaged to obtain a global value of the TWA amplitude (A_{CM}) in the ECG.

Summarizing, given an ECG segment, a TWA episode is characterized by its amplitude (A_{CM}), expressed in μV and its duration (N_{CM}), a dimensionless quantity indicating the number of alternating beats. In addition, a TWA episode is characterized by the overall TWA magnitude ($MAG_{CM} = A_{CM}N_{CM}$), which is expressed in microvolts being a product of amplitude expressed in microvolts and a dimensionless number of alternating beats. Therefore, the TWA magnitude accounts for both TWA amplitude and duration.

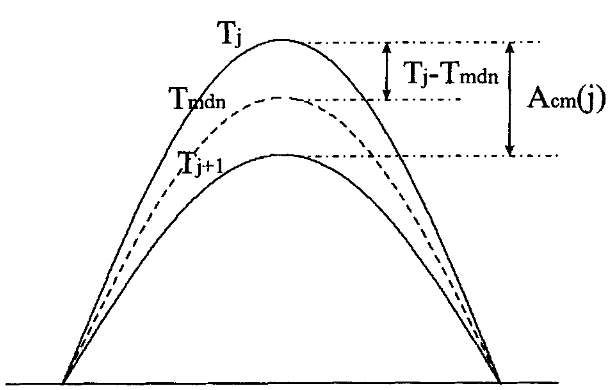

Figure 4. T-wave alternans amplitude is measured between two consecutive T waves, not between a wave and the median T wave (T_{mdn}; thus, $A_{CM}(j) = T_j(i) - T_{j+1}(i)$).

Simulation Protocol

A simulation study was designed to test the ability of our correlation method to detect TWA in simulated 128-beat ECGs. Twelve distinct TWA episodes with various levels of TWA amplitude (μV) and duration (number of alternating beats) were simulated. The duration of a TWA episode indicates the level of its stationarity. Each TWA episode was tested under different "noise" conditions imitating various levels of background noise, baseline wander, respiratory modulation, and RR variability.

The levels of simulated TWA amplitude were 10, 30, 50, and 70 μV, while the duration of the TWA episodes were 32, 64, and 128 beats. Each possible combination of TWA amplitude and duration represented a TWA episode of distinct TWA magnitude. The simulated ECGs had an average RR interval varying between 600 and 1250 ms, and simulated physiological variations of the RR interval between 0% and ± 7.5% of the mean RR. The baseline and the respiratory modulation were simulated as sinusoidal waves, having amplitude and period of 0-150 μV and 3-7 beats, respectively. Uniform white noise of 0-10 μV in amplitude was randomly added to all simulated ECGs. The level of each noise factor (RR variability, baseline wanders, respiratory modulation, and background noise) was randomly and independently assigned to each simulated ECG tracing. Once they have been assigned, the noise factors would stay constant along a specific tracing, but would change going from one tracing to another. Fifteen simulated ECGs, characterized by the same TWA amplitude and duration, but by different levels of the noise factors, were used to represent each TWA episode.

The overall TWA magnitude (MAG_{TWA}) of each simulated TWA episode was computed as the product of the TWA amplitude and duration. We compared the simulated and the estimated TWA magnitudes when using the correlation method (MAG_{CM}) and determined the relative errors ($RE\_MAG_{CM} = |MAG_{TWA} - MAG_{CM}|/MAG_{TWA}$).

Clinical Study Protocol

To test our preprocessing and TWA detection algorithms in clinical setting, we analyzed digital Holter ECG recordings of 39 LQTS patients (RR = 1.05 ± 0.22 s, QT_c = 0.51 ± 0.05 s, age = 22 ± 15 years) from the International Long QT Syndrome Registry[17] and of 36 healthy subjects (RR = 0.90 ± 0.15 s; QT_c = 0.41 ± 0.02 s, age = 39 ± 12 years). For each subject, a 3-channel (X, Y, and Z) Holter ECG was obtained using Burdick Altair Disc recorders (Burdick Inc., Milton, WI, USA) that are able to acquire ECG data at 1000 samples per second per channel for 20 consecutive minutes. To make the recording conditions uniform among patients, all recordings were performed in morning hours and the subjects were asked to lie down and stay still (without movement) during the ECG acquisition. Subsequently, from each 20-minute ECG recording, we randomly selected one 128-beat ECG segment characterized by a good quality of the signal and the absence of cardiac arrhythmias. There was no bias in selecting the ECG segment because we could not identify visually TWA episodes in studied subjects. The 128-beat series was chosen to make our novel approach comparable with the previously reported spectral method.[8] The TWA analysis algorithms were applied to the chosen 128-beat ECG segment.

The control group of healthy subjects was used to define quantitative criteria for TWA detection. The threshold for TWA magnitude was empirically defined by requiring that 97% of the healthy subjects belong to the TWA negative group.

RESULTS

Validation of the Correlation Method in Simulation Studies

We tested the performance of the preprocessing and TWA detection algorithms by introducing randomly added noisy factors, imitating varying levels of background noise, baseline wondering, respiratory modulation, and RR variability (as explained in the Methods section). The overall magnitudes of simulated TWA and the value of the TWA magnitude quantified by the correlation method from simulation are shown in Table 1. The correlation method was able to successfully detect and quantify the sustained (stationary: 128 beats long) and transient (nonstationary: 32-64 beats long) TWA episodes in all tested conditions, with the exception for a short (32 beats long) TWA episode characterized by low amplitude level (10 μV), which was comparable to the simulated background noise level. The median value of relative error for the 12 different simulated conditions of varying amplitude and duration was 31%. Low amplitude (10-30 μV) TWA was detected with higher relative error (mean ± SD = 49 ± 26%) than high amplitude (50-70 μV) TWA (mean ± SD = 16 ± 14%). Thus, in the presence of noisy conditions, the correlation method detects both low and high amplitude TWA, but quantifies the latter with a greater accuracy (P = 0.013). This observation emphasizes the importance of good signal-to-noise ratio required for adequate TWA detection.

Table 1. Quantification of T-Wave Alternans Magnitude Using the Correlation Method in 12 Different Simulated Conditions of Varying T-Wave Alternans Amplitude and Duration*

No. Alternating Beats†	TWA	TWA Amplitude (μV) 10	30	50	70
		TWA Magnitude (μV)			
32/128	Simulated	320	960	1600	2240
(Transient)	QuantifiedbyCM	0 (100%)	696 (28%)	2130 (33%)	2250 (0%)
64/128	Simulated	640	1920	3200	4480
(Transient)	QuantifiedbyCM	392 (39%)	1118 (42%)	3300 (3%)	3968 (11%)
128/128	Simulated	1280	3840	6400	8960
(Sustained)	QuantifiedbyCM	714 (44%)	2303 (40%)	5170 (19%)	6372 (29%)

TWA magnitude = product of TWA amplitude and duration. Each value of the estimated TWA magnitude represents the median of 15 measurements performed in varying 'noisy' conditions (for more explanation, see simulation protocol described in the Methods section). TWA = T-wave alternans; CM = correlation method.
* The values of the relative errors are reported in parentheses.
† = number of alternating beats in a 128-beat series.

T-Wave Alternans Detection By the Correlation Method in Digital Holter Recordings of LQTS Patients

Healthy subjects were used to define criteria for TWA detection. We required that 97% of 36 healthy subjects belong to the TWA-negative group. Consequently, TWA was detected when TWA magnitude (product of the TWA amplitude and duration) was above 550 μV. Such value corresponds to detectable TWA amplitude of at least 4 μV when the TWA episode has a duration of 128 beats. When analyzing TWA in 39 patients with LQTS, the TWA magnitude was above the threshold of 550 μV in 17 patients, who were therefore identified as TWA-positive (Fig. 5). In Table 2, we report the TWA parameters for those TWA-positive LQTS patients.

The TWA-positive LQTS patients were classified according to the three levels of nonstationarity (similarly to simulation study): number of alternating beats < 32, 32-64, and > 64, respectively (Table 3). Eight of the 17 patients with detected TWA had transient (nonstationary) TWA (number of alternating beats $\leq$ 64), with one patient having a very short TWA episode (number of alternating beats = 21), and 7 patients having a mean TWA duration of 53 $\pm$ 11 beats.

The influence of RR duration and beat-to-beat RR variability of preceding beats on TWA magnitude was evaluated to determine if sinus RR fluctuations contribute to TWA. No significant association was found between the TWA magnitude and preceding RR duration or RR interval variability in studied patients ($r < 0.10$; NS).

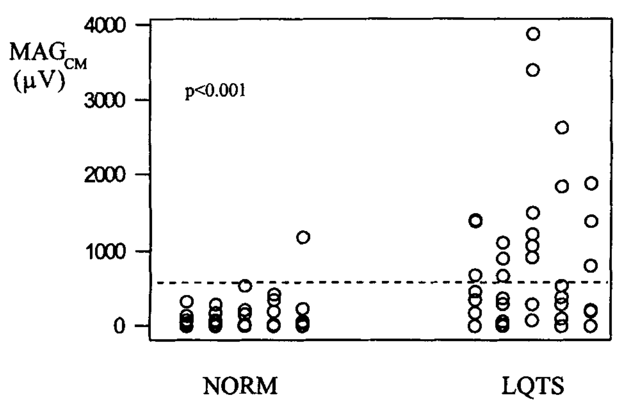

Figure 5. TWA magnitude estimated by the correlation method for healthy subjects (NORM) and long QT syndrome patients (LQTS). The horizontal line (at 550 μV) shows the threshold defined as 97 percentile for healthy subjects.

Table 2. T Wave Alternans (TWA) Parameters in TWA-Positive LQTS Patients

TWA (+) Patient	N_{CM}	A_{CM} (μV)	MAG_{CM} (μV)
1	68	10	680
2	97	40	3880
3	54	26	1404
4	70	20	1400
5	79	14	1106
6	66	14	924
7	51	11	561
8	58	26	1506
9	62	13	806
10	47	30	1410
11	21	51	1071
12	79	24	1896
13	65	14	910
14	64	53	3392
15	72	17	1224
16	33	56	1848
17	91	29	2639

N_{CM} = number of alternating beats quantified by correlation method; A_{CM} = amplitude of T-wave alternans quantified by correlation method; MAG_{CM} = magnitude of T-wave alternans quantified by correlation methods.

DISCUSSION

Microvolt T-wave alternans is more likely to occur during tachycardia, and therefore cardiac pacing or exercise testing have been proposed and clinically used when quantifying TWA.[8,9] Because Holter ECG recordings provide long-term ECG data and episodes of tachycardia are frequently recorded during a 24-hour ambulatory ECG monitoring, we designed a computer algorithm to detect and quantify TWA during Holter-recorded sinus rhythm. Several factors may compromise TWA detection: spontaneous RR variability, muscular or electrical noise, baseline wandering, and respiratory modulation. To limit the influence of these unwanted conditions on TWA detection, we developed a complex preprocessing algorithm consisting of several signal-processing steps (low-pass filtering, RR variability screening, baseline wander and respiratory modulation removal, T-wave windowing and synchronization) able to remove or diminish the influence of the previously mentioned factors. Our preprocessing algorithm was successfully tested in our previously reported simulation experiments that tested the influence of single disruptive

Table 3. T-Wave Alternans (TWA) Stationarity by Correlation Method in LQTS Patients

TWA Stationarity (No. TWA Beats)	No. Patients	TWA Amplitude (μV)	No. Alternating Beats	TWA Magnitude (μV)
No TWA	22	—	—	—
Transient TWA (<32 beats)	1	51	21	1071
Transient TWA (32–64 beats)	7	31 ± 18	53 ± 11	1561 ± 918
Sustained TWA (>64 beats)	9	20 ± 9	76 ± 11	1629 ± 1035

factors on TWA detection.[13] In this study, we confirmed the ability of our new correlation method to detect TWA in a complex simulation protocol (random combinations of the various disruptive conditions) that showed that the correlation method provides a powerful technique to detect and quantify TWA in the presence of noise conditions, as may occur in resting ECGs.

The detection of nonstationary TWA is of particular importance since Holter recordings with visible T-wave alternans (Fig. 1) show that TWA episodes are transient and frequently of short duration. The other methods currently used to detect TWA assume TWA to be stationary or slowly varying. In particular, the spectral method, usually applied on 128-beat ECGs, assumes TWA to be present during the entire ECG, and thus provides an averaged measurement of the TWA amplitude. By definition, the spectral method has a limited ability to detect nonstationary phenomena because to be reliable any spectral estimation has to involve a large number of points (≥ 64). When detecting TWA, this condition corresponds to analyzing a large number of beats (since each point used to estimate the power spectrum corresponds to a T wave). The complex demodulation[7] partially overcomes this limitation, being able to identify TWA in 30 beats or more. However, a potential limitation of the widespread use of the complex demodulation for evaluating TWA is the intrinsic sophistication of the analysis, which requires special software and hardware.[7]

These observations lead us to the concept of a new TWA detection algorithm, the correlation method, able to quantify not only the amplitude, but also the length of TWA episodes. Because the time domain is typically more appropriate for nonstationary signals, our algorithm is time-domain based and computes an alternans index for each consecutive beat. Such index expresses the similarity of each wave to a reference one. Oscillations with a 2:1 pattern in the value of the alternans correlation index reflects the presence of TWA. We arbitrarily chose to detect TWA when there were at least seven consecutive alternating beats. The clinical significance of short episodes of TWA is yet to be investigated since the clinical significance attributed to TWA detected by the spectral method is not sufficient to claim prognostic significance of transient TWA episodes. We believe that it is important and clinically justified to provide a signal-processing technique able to detect both stationary and nonstationary TWAs and therefore to distinguish between transient and sustained episodes of TWA, the task that cannot be accomplished by the spectral method.

Our simulation study in which various levels of noisy factors were randomly added to the simulated ECGs proved that underestimation of TWA magnitude (median value of relative error: 31%) is the limitation of the correlation method. However, most algorithms for TWA detection, including the spectral method, share this limitation, because they are based on the assumption that all samples of the repolarization segment alternate the same quantity.[13] In ECGs with visible TWA (Fig. 1) it is easily seen that alternans may involve only part of the repolarization segment (for example T-wave apex area). However, this assumption cannot be removed because, especially in the case of microvolt TWA, it is not possible to know a priori what part of the T wave is alternating. The simulation study reported in this article also emphasizes the importance of a good signal-to-noise ratio that should be achieved by maintaining good quality of the ECG recording for TWA detection purposes.

To test our method in clinical conditions we analyzed high resolution Holter ECG recordings of 39 LQTS patients, because they are particularly prone to develop TWA.[4] Among the 39 LQTS patients undergoing 3-channel (X, Y, and Z leads) Holter ECG recordings, TWA was identified in 17 patients. Investigating features of TWA, we found that 8 (47%) of the patients had nonstationary TWA (number of alternating beats $\leq$ 64), with 1 of the TWA-positive patients having < 32 beats alternating and 7 patients having 32-64 alternating beats. The observation that many patients with detectable TWA during sinus rhythm had nonstationary TWA further supports the concept of our correlation method able to detect TWA in as few as 7 beats. The spectral method, which assumes constant TWA amplitude over 128 consecutive T waves, is likely to miss nonstationary TWA unless the amplitude of TWA is large enough.

In the 17 patients in whom TWA was detected, no association was observed between TWA magnitude and duration of preceding RR, and between TWA magnitude and beat-to-beat RR variability (RR alternans). This observation provides further evidence for a feasibility of TWA detection during sinus rhythm and for a robust performance of our preprocessing algorithm.

We provide evidence that our new correlation method with appropriate preprocessing of ECG signal can detect nonstationary (transient) episodes of microvolt TWA. Our clinical data showing that about half of TWA-positive long QT syndrome patients have nonstationary TWA emphasize the need for a methodology able to detect and quantify transient TWA episodes. In light of the previously mentioned observations, we believe that our new correlation method for TWA detection during sinus rhythm recorded in standard Holter ECG monitoring will provide a useful and practical tool for identifying patients at risk for life-threatening ventricular arrhythmias.

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